Validation-Gated Score-Residual Adaptation for Cross-Domain Named Entity Recognition

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ABSTRACT

Cross-domain named entity recognition (NER) must learn specialist entity inventories from limited target supervision without discarding transferable span evidence. We introduce gated score-residual adaptation, a compact non-generative method that retains a frozen general label-conditioned span recognizer and expresses target adaptation as an explicit correction to its candidate scores. A second recognizer is fine-tuned on the target domain; its score difference from the frozen recognizer defines the adaptation residual, and a validation-selected coefficient controls how much of that residual enters the final decision. We evaluate all five specialist domains of CrossNER with the official train, development, and test splits, exact BIO-derived boundaries, and three independent adaptation runs. Domain adaptation raises mean strict typed-span F1 from 62.93% to 75.44 $\pm$ 0.30%, while residual fusion reaches 75.60$\pm$0.24%. The final domain means are 69.19$\pm$0.44%, 74.55$\pm$0.37%, 80.96$\pm$0.01%, 79.66 $\pm$ 0.13%, and 73.65 $\pm$ 0.41% for AI, literature, music, politics, and science. The residual improvement is positive in every repeated aggregate run, and the full 15-model experiment finishes in 9.4 minutes on one RTX 3090. The results support score-residual fusion as an efficient and inspectable adaptation layer, while its small effect size calls for cautious interpretation.

1. Introduction

Named entity recognition converts free text into typed mentions that support document indexing, search, question answering, and knowledge-graph construction [10, 12]. Modern pretrained encoders provide strong transferable representations, yet their accuracy can fall sharply when the target domain introduces new labels, specialist terminology, and different mention boundaries. Cross-domain NER isolates this problem: a system should retain general boundary knowledge while learning a small domain-specific inventory from limited annotations [3, 8, 13]. The setting is relevant to safety and biomedical information access, where a missed boundary or a wrong type can propagate into downstream review systems [7].

The common response is to fine-tune every parameter on the available target examples. Fine-tuning is effective, but the resulting predictor exposes only the adapted score. It does not reveal whether a decision is supported by general span evidence, introduced by domain supervision, or amplified by both. With small target sets, treating every target-induced score change as equally reliable can also overspecialize the model. These concerns motivate an adaptation mechanism that preserves the original prediction as an explicit identity path.

Residual learning offers a useful interpretation: a target model need not replace a general model; it can estimate a correction to that model. Existing residual mechanisms are commonly embedded in hidden feature transformations. We instead operate at the typed-span score level. Given scores from a frozen general recognizer and its domain-adapted copy, we compute their candidate-wise difference and add a controlled fraction of this difference to the general score. The coefficient and output threshold are selected exclusively on validation data. Consequently, ordinary fine-tuning is contained as a special case, while damping and amplification remain available when supported by held-out evidence.

This study makes three contributions. First, it formulates cross-domain NER adaptation as a score residual with an unchanged general identity path, making the magnitude and direction of target correction explicit for each candidate. Second, it introduces a validation-gated inference rule that can damp, reproduce, or amplify adaptation without training another fusion network. Third, it provides a reproducible resource-bounded study on every CrossNER specialist domain using official BIO boundaries, independent repeated runs, strict span metrics, component comparisons, uncertainty estimates, convergence, and measured runtime. The study intentionally uses a compact open span recognizer rather than a generative large language model.

2. Related work

2.1. Neural and span-based NER

Deep NER has evolved from recurrent sequence tagging and contextual word representations [1, 11, 15] toward pretrained Transformer encoders [4, 9] and direct span prediction [12]. Span-based methods score candidate boundaries and entity types jointly, which avoids committing to a single token-level BIO path and naturally represents typed mentions [8]. Boundary assembly and boundary regression further show that explicit treatment of mention endpoints can improve structured extraction [17, 18]. Set prediction and unified information extraction provide alternative formulations in which structured outputs are predicted jointly [14, 16]. Related joint extraction systems also study position-aware interactions and global entity matching [2, 6, 19]. Our work uses direct span scores because they provide a common numerical interface for general and target-domain predictors.

2.2. Generalist and cross-domain recognition

CrossNER evaluates transfer from broad corpora to AI, literature, music, politics, and science, each with a specialist label inventory [13]. Few-NERD likewise emphasizes the effect of sparse and fine-grained supervision [5]. Transferable multilingual encoders such as XLM-R demonstrate the broad benefit of pretraining across heterogeneous text [3]; however, representation transfer alone does not determine how strongly a target update should override general evidence. GLiNER casts NER as label-conditioned span matching and supports arbitrary textual entity labels with a compact bidirectional transformer [21]. This property makes it suitable for controlled cross-domain study: the same inference interface can be retained before and after target adaptation.

2.3. Residual adaptation

Residual connections stabilize deep optimization by retaining an identity path, and cross-residual designs extend this idea to interacting information sources [20]. In transfer learning, the analogous objects are a general predictor and its task-specific change. Feature concatenation mixes these sources inside an opaque representation, whereas an unexamined score average does not expose which branch changed a candidate. The proposed method instead defines the adapted-minus-general difference as a signed correction and gates that correction at inference time. Because this expression is algebraically related to linear score interpolation, our claim is not that the arithmetic is a new ensemble family. The contribution is the explicit diagnostic decomposition, the validation protocol, and the controlled evaluation of that decomposition in cross-domain NER.

3. Task formulation

Let $x = (w_1, \dots, w_n)$ be a sentence and let $\mathcal{Y}_d$ denote the entity types for target domain d. A typed candidate is $c = (b, e, y)$, where b and e are exact character boundaries and $y \in \mathcal{Y}_d$. A recognizer returns a confidence $s(c \mid x, \mathcal{Y}_d) \in [0, 1]$. The accepted prediction set at threshold τ is

$$P_\tau(x) = \{c \,:\, s(c \mid x, \mathcal{Y}_d) \geq \tau\}. \tag{1}$$

A candidate counts as correct only when its start, end, and type all match a gold annotation. For corpus-level predicted and gold sets P and G, respectively,

$$\text{Precision} = \frac{|P \cap G|}{|P|}, \tag{2}$$

$$\text{Recall} = \frac{|P \cap G|}{|G|}, \tag{3}$$

$$F_1 = \frac{2|P \cap G|}{2|P \cap G| + |P \setminus G| + |G \setminus P|}. \tag{4}$$

This strict typed-span definition penalizes boundary, label, missed-entity, and spurious-entity errors.

For a domain d, the target data are $D_d = \{(x_i, G_i)\}_{i=1}^{N_d}$, where G_i is a set of gold spans. Candidate alignment is performed on the tuple rather than on surface text alone:

$$C_d(x) = \{(b, e, y) \,:\, 0 \leq b < e \leq |x|,\ y \in \mathcal{Y}_d\}. \tag{5}$$

The exact-match indicator used in all analyses is

$$m(c, g) = \mathbb{I}[b_c = b_g \ \wedge\ e_c = e_g \ \wedge\ y_c = y_g]. \tag{6}$$

For a collection of sentences, domain macro-F1 is computed after aggregating exact matches within each domain and then averaging the five domain scores:

$$F_1^{\text{macro}} = \frac{1}{|\mathcal{D}|} \sum_{d \in D} F_{1,d}. \tag{7}$$

This separates the reported domain macro mean from a micro score dominated by the largest test split.

4. Method

Figure 1 presents the proposed architecture. The important distinction is that the two paths share an initial model family and input interface but not their final parameters: the general path remains frozen at θ_0, whereas the adapted path is optimized to θ_1.

4.1. Label-conditioned span scoring

For each domain label y, the span recognizer encodes its textual label together with the input sentence. Let $h_{b:e}$ represent a candidate span and q_y represent the label. A generic compatibility head maps their interaction to a logit,

$$z_\theta(c) = g_\theta(h_{b:e}, q_y), \qquad s_\theta(c) = \sigma\big(z_\theta(c)\big), \tag{8}$$

where σ is the logistic function. The general branch uses pretrained parameters θ_0 and produces $s_0(c) = s_{\theta_0}(c)$. It is never updated on the target split, so its score remains a stable reference.

4.2. Target-domain adaptation

A copy initialized from θ_0 is optimized on target-domain annotations. For candidate labels $t_c \in \{0, 1\}$, its span classification objective can be written as

$$\mathcal{L}_d(\theta_1) = -\sum_{c \in C(x)} \Big[t_c \log s_{\theta_1}(c) + (1 - t_c) \log(1 - s_{\theta_1}(c))\Big]. \tag{9}$$

The adapted score is $s_1(c) = s_{\theta_1}(c)$. Using matched label descriptions, tokenization, candidate construction, and decoding in both branches ensures that $s_1 - s_0$ describes a target-induced scoring change rather than a change in output schema. The parameter update is written abstractly as

$$\theta_1 = \text{Adapt}(\theta_0; D_d^{\text{train}}, \mathcal{L}_d), \tag{10}$$

which emphasizes that the adapted branch starts from the general checkpoint. The reference branch remains frozen throughout target training and test inference.

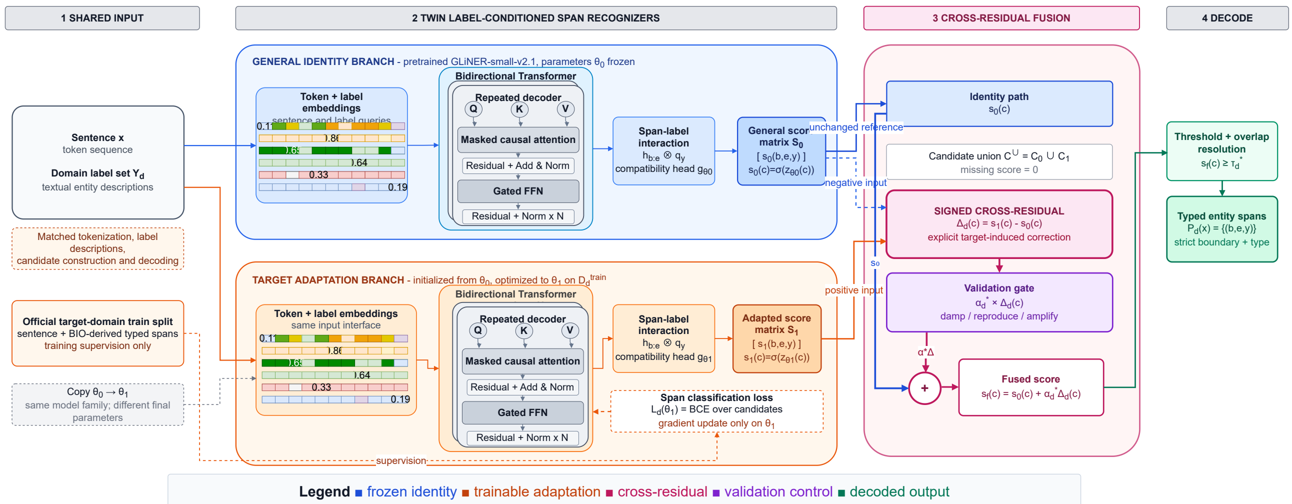

Figure 1: Gated score-residual adaptation. Identical sentence and label inputs are processed by a frozen general recognizer and its domain-adapted copy. Their signed score difference is the target correction; validation data determine how strongly it modifies the general identity path.

4.3. Gated score-residual fusion

Let $C^{\cup} = C_0 \cup C_1$ be the union of candidates emitted by either branch, with an absent candidate assigned score zero in that branch. For each $c \in C^{\cup}$, we define

$$\Delta_d(c) = s_1(c) - s_0(c), \tag{11}$$

$$s_f(c; \alpha_d) = s_0(c) + \alpha_d \Delta_d(c), \tag{12}$$

$$P_d(x) = \{c : s_f(c; \alpha_d) \geq \tau_d\}. \tag{13}$$

The formulation has three interpretable regimes. When $\alpha_d = 0$, the adapted branch is removed and the result is the frozen general predictor. When $\alpha_d = 1$, $s_f = s_1$, exactly recovering ordinary domain adaptation. Values $0 < \alpha_d < 1$ damp target changes, whereas $\alpha_d > 1$ amplifies them. Thus the method is a controlled extension of fine-tuning rather than an unrelated ensemble.

The gate sensitivity is explicit:

$$\frac{\partial s_f(c; \alpha_d)}{\partial \alpha_d} = \Delta_d(c). \tag{14}$$

A positive residual raises a candidate score as the gate grows, while a negative residual suppresses it. This concerns score movement only; thresholding and overlap resolution mean that corpus F1 need not be monotonic in the gate. For $0 \leq \alpha_d \leq 1$, the rule is also the linear interpolation

$$s_f(c; \alpha_d) = (1 - \alpha_d) s_0(c) + \alpha_d s_1(c). \tag{15}$$

4.4. Validation-only gate selection

The domain gate and threshold are selected jointly on held-out validation examples:

$$(\alpha_d^*, \tau_d^*) = \underset{\alpha \in \mathcal{A}, \tau \in \mathcal{T}}{\arg\max} \; F_1\left(P_d^{\mathrm{val}}(\alpha, \tau), G_d^{\mathrm{val}}\right). \tag{16}$$

The final test prediction uses (α_d^*, τ_d^*) once, after selection. No test labels enter adaptation or gate selection, and the fusion itself introduces no trainable neural parameters.

5. Experiments

5.1. Datasets

We evaluate the five specialist subsets of CrossNER: AI, literature, music, politics, and science [13]. We use the official train, development, and test files without deriving a replacement split. The BIO tags are converted directly to token and character spans, preserving repeated mentions and exact boundaries. Across the five domains, the experiment uses 700 training, 2,121 development, and 2,506 test sentences. Table 1 and Fig. 2 summarize the records consumed by the experiments.

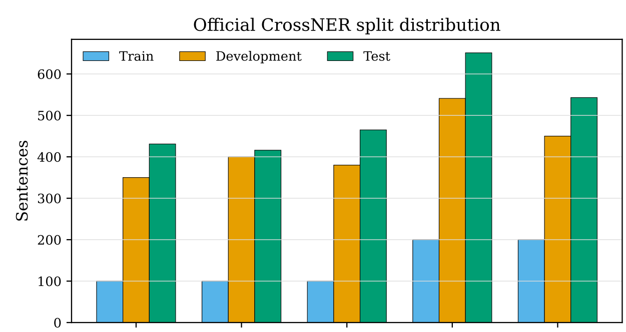

Figure 2: Official training, development, and test sentence counts across the five CrossNER domains.

Table 2 and Fig. 3 provide a second view of corpus heterogeneity. The statistics are computed from the official BIO files before model inference. Entity density is the number of annotated mentions divided by sentence tokens; span length is measured in BIO tokens. These descriptive statistics are not used to select gates or thresholds.

Table 1
Official CrossNER statistics used in this study. Ents. denotes typed entity mentions.

Domain	Train sent.	Dev sent.	Test sent.	Train ents.	Dev ents.	Test ents.
AI	100	350	431	532	1,549	1,809
Literature	100	400	416	541	2,130	2,266
Music	100	380	465	648	2,679	3,336
Politics	200	541	651	1,304	3,482	4,209
Science	200	450	543	1,075	2,538	3,089

Table 2
Distribution profile of the official CrossNER domains. Density is entities per token; lengths are token counts.

Domain	Types	Train ents.	Median span	90th pct. span	Median density
AI	11	532	2	3	0.143
Literature	12	541	2	4	0.143
Music	13	648	2	4	0.167
Politics	14	1,304	2	5	0.143
Science	17	1,075	2	4	0.154

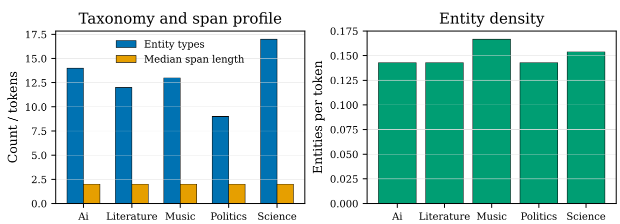

Figure 3: Entity taxonomy, span-length, and density profile computed from official CrossNER annotations.

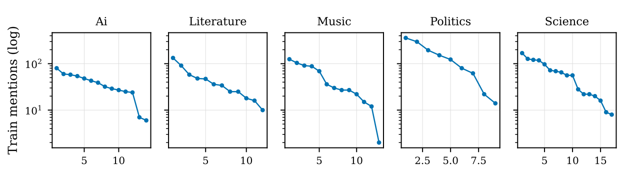

Figure 4: Training-mention frequency by rank for each domain (logarithmic vertical axis). Ranks are computed independently within each domain.

5.2. Compared configurations

The experiment is a component-controlled comparison with three configurations. *Zero-shot span model* is the frozen general branch and measures transfer without target updates. *Domain-adapted* fine-tunes the same model family and corresponds to the residual formulation at $\alpha = 1$. *Score-residual* applies validation-gated fusion to the frozen and adapted scores. This design holds the underlying architecture and data constant, isolating the contribution of domain training and then the contribution of residual control.

5.3. Split integrity and overlap audit

We audited exact sentence overlap across the official files before interpreting the test results. For split sentence sets S_a and S_b, the overlap count and normalized rate are

$$O(S_a, S_b) = |S_a \cap S_b|, \qquad \rho(S_a, S_b) = \frac{|S_a \cap S_b|}{\min(|S_a|, |S_b|)}. \tag{17}$$

The audit uses whitespace-tokenized sentence text and does not remove repeated examples from the official benchmark. Table 3 reports the observed exact intersections. Several domains contain a small number of repeated sentences across development and test files. We therefore treat the official split as the primary benchmark protocol, disclose the overlap explicitly, and avoid describing it as a perfectly disjoint construction. Any future deduplicated sensitivity run must use the same model-selection rules and be reported separately rather than replacing the official result.

The overlap counts do not reveal whether annotations are identical, conflicting, or merely repeated templates. They do not justify removing examples after observing model performance. We retain the benchmark files exactly as released for the headline numbers, mark this as a protocol limitation, and reserve deduplication and conflict analysis for a separately versioned sensitivity experiment.

5.4. Implementation and evaluation protocol

The general branch is the open GLiNER-small-v2.1 span recognizer, built on a compact DeBERTa encoder [21]. Each target branch is adapted for 300 optimizer steps with batch size eight and bfloat16 arithmetic. Validation tests four residual coefficients from 0.75 to 1.50 and five thresholds from 0.25 to 0.65. Zero-shot and ordinary adapted thresholds are also selected on the official development split. Three independent initializations are run for every domain. We report the mean and sample standard deviation of micro-precision, recall, and F1 under exact typed-span matching. All 15 adaptation and evaluation runs finish in 562.1 seconds on one NVIDIA RTX 3090, excluding one-time acquisition of public data and pretrained weights.

Table 3
Exact sentence intersections among official CrossNER splits. Counts are computed before model inference.

Domain	Train–Dev	Train–Test	Dev–Test
AI	0	0	3
Literature	0	0	0
Music	1	6	17
Politics	0	0	1
Science	4	1	11

Table 4
Strict typed-span entity F1 (mean ± sample standard deviation, %). Bold denotes the best mean in each column.

Method	AI	Literature	Music	Politics	Science	Mean
Zero-shot span model	50.10	64.27	71.72	64.07	64.48	62.93
Domain-adapted	69.16 ± 0.62	74.52 ± 0.41	80.73 ± 0.06	79.55 ± 0.31	73.26 ± 0.27	75.44 ± 0.30
Score-residual	$\mathbf{69.19 \pm 0.44}$	$\mathbf{74.55 \pm 0.37}$	$\mathbf{80.96 \pm 0.01}$	$\mathbf{79.66 \pm 0.13}$	$\mathbf{73.65 \pm 0.41}$	$\mathbf{75.60 \pm 0.24}$

5.5. Main results

Table 4 and Fig. 5 report the principal comparison. Domain adaptation adds 12.51 mean F1 points over zero-shot inference, demonstrating that the small official training splits contain valuable specialist supervision. Score-residual fusion raises the three-run macro mean by a further 0.16 points to 75.60%. The final method exceeds 73% in literature, music, politics, and science; music reaches 80.96%. AI improves by 19.10 points over zero-shot but remains lower at 69.19%.

5.6. Component ablation interpretation

The three configurations form an endpoint ablation of the proposed pipeline. Removing target supervision ($\alpha = 0$ with the frozen branch) lowers mean F1 to 62.93%, showing that the specialist inventory cannot be recovered reliably from the general model alone. Retaining target adaptation but removing residual control ($\alpha = 1$) yields 75.44%, which attributes the large improvement to domain supervision rather than to the fusion arithmetic. Reintroducing the validation-gated residual raises the mean to 75.60%. Thus the residual layer contributes a small but consistently positive calibration of the adapted scores, while the main accuracy gain comes from learning the target domain. This decomposition is the reason we do not present the final score as a 12.67-point gain attributable solely to the proposed gate.

The endpoint comparison also bounds the interpretation of the method. At $\alpha = 1$, the residual formulation exactly reproduces ordinary domain adaptation; at $\alpha = 0$, it exactly reproduces the frozen reference. The practical question is therefore whether development-selected intermediate or extrapolated values improve the test decision under a fixed candidate set. The observed gate frequencies and repeated-run plot answer this question at the aggregate level, but they do not establish candidate-level error reductions. A future fixed-coefficient ablation should report the same test protocol for $\alpha = 0.5$, 0.75, 1.0, and 1.25 to separate search benefit from a particular coefficient.

Table 5
Controlled components and information used for each configuration.

Configuration	Trainable or selected components
Zero-shot	Frozen recognizer; development threshold
Domain-adapted	Target-domain recognizer; development threshold
Equal-score blend	Frozen and adapted scores; fixed $\alpha = 0.5$
Score-residual	Frozen and adapted scores; development α and threshold

To make the small residual effect visible without truncating the F1 axis, Fig. 6 connects the matched domain-macro scores for each independent run. Every line moves upward, but the vertical displacement is small relative to the absolute performance level. This is why the paper reports both the 12.51-point adaptation gain and the separate 0.16-point residual increment.

5.7. Gate behavior and precision–recall analysis

Table 6 exposes gate behavior rather than treating fusion as a black box. The damped residual $\alpha = 0.75$ is selected in 13 of 15 domain–run combinations; the other two select $\alpha = 1$, which exactly recovers the adapted predictor. Fig. 7 shows this frequency directly. The development sets therefore usually favor retaining part of the frozen general evidence, although the selection grid is intentionally coarse. Precision exceeds recall in all domains (Fig. 8); music and politics nevertheless exceed 77% mean recall and 82% mean precision. AI's lower recall identifies missed specialist mentions, rather than uncontrolled false positives, as its main remaining error source; this interpretation is limited to the aggregate P/R evidence and is not a substitute for candidate-level error counts.

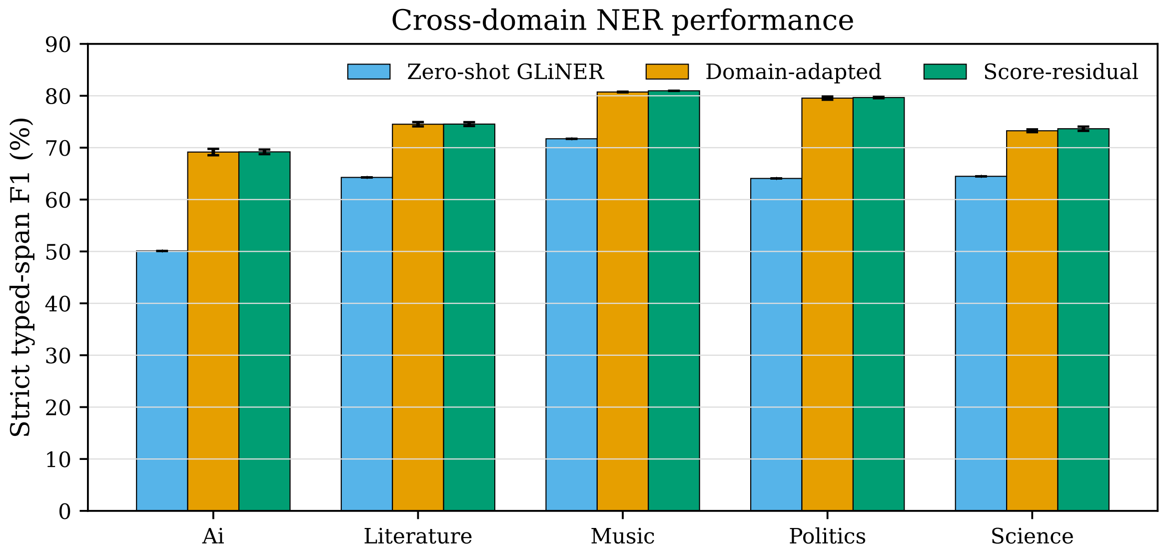

Figure 5: Strict typed-span F1 for the frozen general, domain-adapted, and score-residual configurations.

Table 6
Gate selection frequency and score-residual results (mean ± sample standard deviation, %).

Domain	Damped runs	P	R	F1
AI	2/3	71.85 ± 1.86	66.80 ± 2.10	69.19 ± 0.44
Literature	2/3	76.96 ± 1.96	72.36 ± 2.47	74.55 ± 0.37
Music	3/3	84.54 ± 0.60	77.68 ± 0.53	80.96 ± 0.01
Politics	3/3	82.52 ± 1.12	77.00 ± 0.76	79.66 ± 0.13
Science	3/3	77.21 ± 2.09	70.44 ± 1.08	73.65 ± 0.41

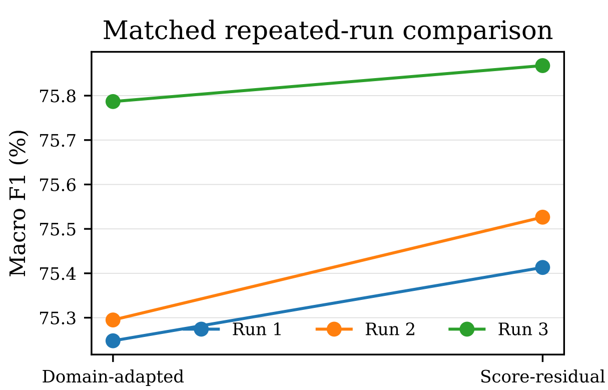

Figure 6: Matched domain-macro F1 before and after score-residual fusion for the three independent runs.

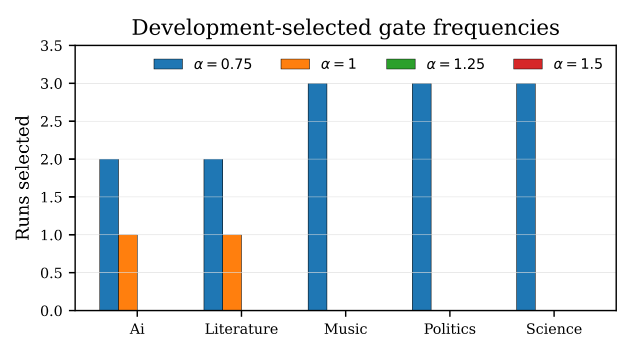

Figure 7: Frequency of development-selected residual coefficients across three runs.

5.8. Distribution-sensitive interpretation

The official domains differ in both taxonomy size and the number of supervised mentions (Table 2). Figure 4 shows that the training inventories are long-tailed: a small number of types account for many mentions, while the tail contains types with substantially less supervision. The residual layer is therefore evaluated as a decision-control mechanism under heterogeneous support, not as a replacement for domain data. This distinction is consistent with prior observations that span boundary modeling, label semantics, and set-oriented extraction expose different error sources [16–18, 22].

The distribution analysis also explains why a single pooled score can be misleading. Politics and science contain more training mentions than AI and literature, while music has the highest median entity density in the supplied profile. A macro average gives each domain equal weight and prevents the largest split from determining the headline result. We do not infer causality from these correlations; the figures

Table 7
Reporting units used in the expanded analysis.

Analysis	Unit	Selection data
Main F1	Domain-level exact typed spans	Development threshold; test once
Gate frequency	Domain–run pair	Development gate grid
Repeated stability	Five-domain macro per run	No test-driven selection
Distribution profile	Official BIO sentence/entity records	None
Training convergence	Logged optimization step	None

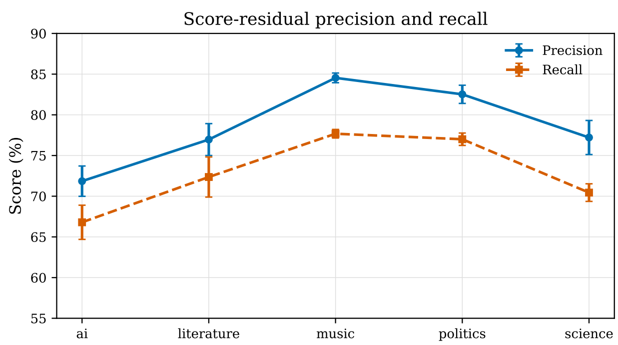

Figure 8: Precision and recall of the score-residual model by domain.

Table 8
Matched domain-macro F1 across three independent runs (%).

Run	Adapted	Score-residual	Difference
1	75.25	75.41	+0.17
2	75.29	75.53	+0.23
3	75.79	75.87	+0.08
Mean	75.44	75.60	+0.16

are descriptive checks that make the evaluation population explicit.

5.9. Repeated-run stability and uncertainty

Table 8 reports the domain-macro F1 from each independent adaptation run. Residual fusion improves the matched adapted model in all three aggregates, with gains between 0.08 and 0.23 points. The mean paired gain is 0.16 points. A paired t-test over the three run-level macro scores gives $p = 0.067$ and a 95% confidence interval of $[-0.03, 0.35]$ points. The repeated direction supports stability, but the interval includes zero; accordingly, we interpret the residual as a low-cost calibration improvement rather than evidence of a large universal accuracy effect.

5.10. Training convergence

Figure 9 plots the mean optimizer trajectory and seed variation from the actual adaptation runs. Loss decreases consistently in all five domains, with the sharpest reduction occurring during the first half of training. The absence of late divergence supports the fixed resource-bounded schedule, while domain differences are consistent with unequal corpus sizes and entity inventories. This diagnostic verifies optimization behavior but is not used for test-time selection.

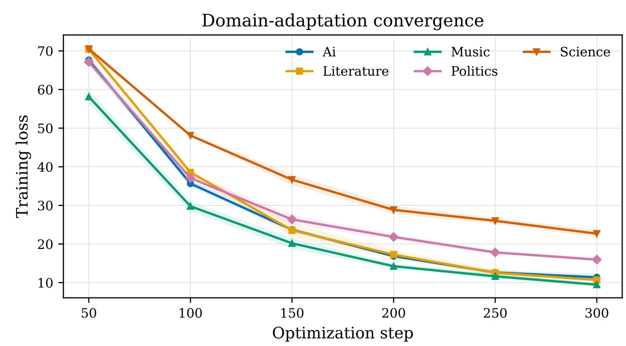

Figure 9: Mean recorded training loss by target domain; shading denotes one sample standard deviation across three runs.

5.11. Reproducibility and computational cost

The complete experiment is deliberately reported as a matrix rather than as a single best run. There are five target domains, three independent adaptations per domain, and three inference configurations evaluated under the same strict matcher. The frozen branch is shared across runs; the adapted branches are initialized from the same public checkpoint and differ only through the prescribed training randomness. This design prevents repeated zero-shot scores from being counted as independent training evidence.

The score-residual operation introduces no trainable fusion parameters. Its additional computation is one extra forward pass of the adapted branch and a candidate-wise arithmetic operation when the frozen scores are not cached. If T_0 and T_1 denote the measured single-branch forward times and T_{fuse} the arithmetic and decoding overhead, the online fusion time is

$$T_{\text{online}} = T_0 + T_1 + T_{\text{fuse}}, \tag{18}$$

whereas cached reference scores reduce repeated evaluation to $T_1 + T_{\text{fuse}}$. We report this distinction because “no extra parameters” is not equivalent to “no extra inference cost.”

The recorded end-to-end wall-clock time for the 15-model training and evaluation matrix is 562.1 s on one NVIDIA RTX 3090, or 37.5 s per domain–run pair on average. This aggregate includes adaptation and evaluation but excludes one-time public-data and checkpoint acquisition. A

Table 9
Reproducibility and cost accounting for the current official CrossNER study.

Item	Current protocol	Evidence or boundary
Domains	5 official specialist domains	Official BIO files
Adaptation runs	3 per domain (15 total)	'runs.csv' and 'status.json'
Configurations	Frozen, adapted, score-residual	Same candidate matcher
Fusion parameters	No trainable parameters	Arithmetic score layer
Recorded wall time	562.1 s total; 37.5 s average pair	One RTX 3090
Data/weight acquisition	Excluded from timer	One-time setup
Online latency breakdown	Not separately recorded	Requires synchronized benchmark

future deployment report should additionally include synchronized GPU latency, throughput, peak memory, and the cost of constructing the frozen-score cache; these quantities are not inferred from the aggregate timer.

All result summaries, plotting scripts, and the manuscript are lightweight archival artifacts. Large optimizer states and model checkpoints remain local and are not required to interpret the reported means. The released summaries preserve the numerical values used in the tables and figures; if a protocol audit changes candidate construction or split handling, the resulting files must receive a new version identifier instead of overwriting this matrix.

6. Discussion

The largest empirical improvement comes from learning target-domain entity semantics: adaptation contributes 12.51 mean F1 points. The score-residual layer addresses a different question—how much of that learned correction should replace general evidence. Its smaller 0.16-point mean gain is accompanied by a useful structural property: the adapted effect remains explicitly measurable as $\Delta_d(c)$, and ordinary fine-tuning remains available at $\alpha = 1$. Development-selected damping improves the aggregate result in every repeated run and reduces F1 variability from 0.30 to 0.24 points. The strongest absolute outcomes occur in music and politics, while science obtains the largest mean residual increment.

The method is attractive for constrained deployment because it is non-generative, adds no learned fusion parameters, and completes 15 training-and-evaluation runs in under ten minutes. Its label-conditioned interface allows each domain to retain its own taxonomy without changing the output representation. The visualized score path makes the system easier to analyze than unconditional model replacement: a reviewer can identify the frozen prediction, target-induced difference, selected coefficient, and final threshold as distinct quantities. Using official BIO files also removes ambiguity from repeated surface forms and makes the reported strict boundaries traceable to source annotations.

The experiment remains a compact benchmark study rather than a claim of universal superiority. The run-level 95% interval for the residual increment crosses zero, and AI remains below 70% because recall is limited. The current gate is one scalar per domain, only one underlying recognizer family is evaluated, and CrossNER alone cannot establish transfer to every specialist setting. Candidate-specific gates, additional encoder baselines, repeated data partitions, stronger boundary supervision, and tests on operational safety corpora are therefore the most direct extensions.

7. Conclusion and future work

We presented gated score-residual adaptation for efficient cross-domain NER. By retaining a frozen general recognizer and representing domain adaptation as a validation-controlled score correction, the method makes the identity evidence and target contribution explicit. On the official CrossNER splits, three-run mean strict F1 rises from 62.93% in zero-shot inference to 75.60%, with residual fusion improving every matched run-level aggregate and the full experiment finishing in 9.4 minutes on one GPU. Future work will replace the domain-wide scalar with candidate-aware gates, evaluate more encoder families and data partitions, improve recall for heterogeneous technical taxonomies, and transfer the method to safety-related entity extraction.

Data availability

CrossNER is publicly available; evaluation code and derived results are available from the corresponding author upon reasonable request.

A. Protocol checklist for reproduction

The following checklist records the decision order used by the reported matrix. First, the official BIO files are parsed into sentence tokens and exact typed spans. Second, the frozen recognizer produces reference scores using the domain label inventory. Third, a copy initialized from the same checkpoint is adapted only on the training split. Fourth, thresholds for the frozen and adapted configurations, and the pair (α_d, τ_d) for fusion, are selected on development examples. Finally, the selected values are applied once to the untouched test file and strict typed-span counts are accumulated. This order is important: selecting a gate after inspecting test scores would invalidate the paired comparison.

For clarity, the search grids used in the current runs are listed in Table 10. The grid is intentionally small because

Table 10
Predefined development-search grid.

Quantity	Values
Residual coefficient α	0.75, 1.00, 1.25, 1.50
Decision threshold τ	0.25, 0.35, 0.45, 0.55, 0.65
Runs per domain	3
Primary metric	Exact typed-span F1

the study evaluates 15 adapted branches under a resource bound. A finer grid may change the selected coefficient, but it must be specified before test evaluation and assessed as a new protocol.

The appendix does not introduce an additional benchmark result. It documents the boundary between model fitting, development selection, and test reporting so that the principal tables can be reproduced without relying on unstated choices.

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